

Supply Chain Performance & Risk Analytics System — Executive Summary

Diagnostic across 1,05,834 cleaned records (1,000 vendors · 49,914 purchase orders · 30,000 inventory snapshots · 19,950 shipments · 4,970 returns) · order window Jan 2021 – Dec 2024 · all values INR.

On-Time Delivery	Procurement Spend	Shipment Damage	Return Value	Stockout Rate (latest)	Top-10 Vendor Share
44.3% (vs 85%)	Rs 31,567 Cr	17.7%	Rs 315 Cr	9.96%	1.5%

Situation

The company runs a low-reliability, high-cost supply chain. Delivery performance is roughly half the industry benchmark, the vendor base is large and undifferentiated, working capital is trapped in overstock while service still fails at the margin, and product damage and returns are running at multiples of normal levels. The current vendor Reliability_Score does not explain any of this.

Key findings (data-referenced)

- **Delivery:** only **44.3%** of 46,848 delivered orders were on time vs the 85% benchmark — a 40.7-point gap. 55.7% arrived late, on average 6.6 days (up to 30).
- **Vendor scoring is broken:** correlation between Reliability_Score and delivery delay is **r = 0.00**. Tier 1 vendors are late 56.1% of the time; Tier 3, 55.4% — statistically indistinguishable.
- **Spend is fragmented:** Rs 31,567 Cr across 1,000 vendors; the top 10 are just **1.5%** of spend and all 8 categories sit within 12.3–13.0% — little leverage, high administration cost.
- **Inventory imbalance:** turnover is healthy (16.6×/yr, no slow-moving category), but **24.7%** of latest-snapshot SKU-lines are overstocked vs 10.0% in stockout; WH06 Kolkata is worst on stockout at 11.9%.
- **Logistics:** Rs 249 Cr freight, near-identical average cost across all modes (<7% spread) and a **17.7%** damage rate that is uniform across all 8 carriers — a systemic packaging/handling problem, not a carrier problem.
- **Returns:** Rs 315 Cr on 4,970 lines (9.96% of PO lines); Damaged Packaging, Defective Item and Quality Failure together account for ~42% of return lines.

Top 5 strategic recommendations

- **Launch a delivery-recovery program.** Add contractual OTIF targets and a weekly late-order review for the worst 50 vendors; target +25 points of on-time delivery within two quarters.
- **Rebuild the vendor score on outcomes.** Replace the current rating with an index of actual OTIF, defect rate and return rate from vw_supply_chain_fact; re-tier vendors quarterly.
- **Consolidate the supplier tail.** Move the 238 RFM 'Dormant/At-Risk' and 'Low-Value' vendors (Rs 6,088 Cr) into a smaller set of strategic partners to gain pricing leverage and cut transaction cost.
- **Right-size inventory.** Cut safety stock where overstocked and tighten reorder points at WH06 Kolkata and the four next-worst sites; release working capital without lowering service.
- **Fix packaging and inbound QC.** Fund a packaging redesign and an inbound quality gate to drive the damage rate below 10% and cut the quality/packaging share of returns.

Deliverables: 5 cleaned datasets · Excel cleaning & statistics workbook (Null Summary, Data Dictionary, Descriptive Stats, Correlation, Vendor RFM) · MySQL script (schema + master fact view + 15 commented analytical queries) · 4-page Power BI dashboard specification & reference build · 18-slide presentation · this summary.